Ex. No.: 7

Date:

IPC USING SHARED MEMORY

Aim:

To write a C program to do Inter Process Communication (IPC) using shared memory between sender process and receiver process.

Algorithm:

sender

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Write a string to the shared memory segment using sprintf
5. Set delay using sleep
6. Detach shared memory segment using shmdt

receiver

1. Set the size of the shared memory segment
2. Allocate the shared memory segment using shmget
3. Attach the shared memory segment using shmat
4. Print the shared memory contents sent by the sender process.
5. Detach shared memory segment using shmdt

Program Code:

sender.c

```
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#define SharedMemSize 50
void main()
{
 char c;
 int shmid;
 key_t key;


 char *shared_memory;
```

```
 key = 5677;
 //Create segment with the key specified
 if ((shmid = shmget(key, SharedMemSize, IPC_CREAT |
0666)) < 0)
 {
          //perror explains error code
 perror("shmget");
 exit(1);
 }

 //Attach the segment
 if((shared_memory= shmat(shmid, NULL, 0)) == (char *) -1)   {
 perror("shmat");
 exit(1);
 }

 sprintf(shared_memory," Welcome to Shared Memory");
sleep(2);
 exit(0);
 }
```

receiver.c

```
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <stdio.h>
#include <stdlib.h>
#define SharedMemSize 50

void main()
{
 int shmid;
 key_t key;
 char *shared_memory;
 key = 5677;
 if ((shmid = shmget(key, SharedMemSize, 0666)) < 0)   {
 perror("shmget");
 exit(1);


 }

 //Attach the segment to our data space
 if((shared_memory = shmat(shmid, NULL, 0))==(char *) -1)   {
 perror("shmat");
 exit(1);
 }
 //Read the message sender sent to the shared memory
printf("Message Received: %s \n",shared_memory);  exit(0);
 }
```

Output:

Terminal 1

```
[root@localhost student]# gcc sender.c -o sender
[root@localhost student]# ./sender
```

Terminal 2

```
[root@localhost student]# gcc receiver.c -o receiver
[root@localhost student]# ./receiver
Message Received: Welcome to Shared Memory
[root@localhost student]#
```